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Acoustic-Driven Generation of mFDA Clinical Speech Reports with Flan-T5: Reproduction, Optimisation, and a Study of Evaluation-Metric Comparability

Final Project Report

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Abstract

Clinical speech assessment for Parkinson’s disease (PD) relies on perceptual scales such as the modified Frenchay Dysarthria Assessment (mFDA), which are subjective, expert-intensive, and rarely produce standardised written reports. Building on the reference study of Arias-Vergara et al., this project fine-tunes sequence-to-sequence language models to generate mFDA-style clinical reports directly from seven numeric acoustic biomarkers (breathing, lips, palate, larynx, monotonicity, tongue, and intelligibility). Training uses a large synthetic corpus of simulated biomarker–report pairs; the held-out test set consists of real PD and healthy-control (HC) clinical reports. Through a reproducible extraction–preparation–training–evaluation pipeline and more than twenty controlled experiments, we identify a best configuration, *B17*: a 248M-parameter Flan-T5-base model adapted with rank-32 LoRA. Our central finding is methodological: the apparent gap between our Flan-T5 model and the reference paper’s headline scores is *mostly a measurement artifact*, not a model deficiency. The reference evaluation reports character-level BLEU, whereas standard practice uses word-level BLEU; on the same generated text the two differ by roughly a factor of two. After reproducing the original evaluation exactly—character-level BLEU-2, stemmed ROUGE, and an English BERTScore backbone—B17 attains an average score of **0.649** (character-BLEU 0.759, BERTScore 0.911), placing a 248M model beside the reference paper’s best seven-billion-parameter systems. We further map the Flan-T5-base performance plateau across learning rate, epochs, weight decay, label smoothing, model size, LoRA rank, oversampling, structured targets, and encoder-input format, and we correct a BERTScore language misconfiguration that had systematically undervalued semantic scores. We conclude that B17 is final for the Flan-T5 track and that surpassing the reference headline requires a decoder-only 7B model rather than further small-model tuning.

1 Introduction

1.1 Motivation

Speech is a rich, non-invasive biosignal that reflects many aspects of neurological health. In Parkinson’s

disease (PD), progressive motor impairment frequently produces *dysarthria*—a motor speech disorder caused by weakness, slowness, or poor coordination of the muscles used for speaking. Despite its diagnostic relevance, speech is seldom for-

mally assessed in routine PD evaluation, in part because standardised written reporting protocols are uncommon and because expert perceptual rating is time-consuming and inconsistent between raters.

The modified Frenchay Dysarthria Assessment (mFDA) is one of the few instruments that quantify speech impairment in a structured way. It scores seven perceptual categories—breathing, lips, palate, larynx, monotonicity, tongue, and overall intelligibility—each on an ordinal scale that an expert phoniatrician maps from “normal” to “severe.” While clinically meaningful, the assessment is subjective and demanding: it requires trained listeners and does not directly yield a narrative report a clinician can read, share, or archive.

Two complementary difficulties motivate an automated approach. Purely acoustic analysis produces numeric biomarkers that are objective but hard to interpret without signal-processing expertise; purely perceptual assessment produces interpretable judgements but is labour-intensive. Large language models (LLMs) offer a bridge: given objective acoustic biomarkers as input, a fine-tuned model can emit a structured, human-readable clinical report, combining the objectivity of acoustics with the interpretability of narrative text. Crucially, the acoustic biomarkers act as an explicit, controllable input the clinician can inspect—unlike an end-to-end black box that maps audio directly to text—so the generation remains auditable.

1.2 Task and reference study

This project reproduces and extends the study “Acoustic-Driven Generation of Pathological Speech Reports Using Large Language Models” (Arias-Vergara et al., 2025). The reference work frames report generation as a text-to-text problem: seven acoustic biomarkers, correlated with the seven mFDA categories, are formatted into an input prompt, and a fine-tuned LLM generates an mFDA-style report as output. Because only 50 PD and 50 HC subjects are available—far too few to train a generative model—the authors introduce a data-augmentation strategy that *simulates*

biomarker–report pairs from the statistical distribution of the real features, yielding a large synthetic training corpus. Real subject reports are reserved exclusively for testing, avoiding overfitting to the small clinical set.

The reference study fine-tunes a range of models (T5, Flan-T5, GPT-2, LLaMA-7B, Mistral-7B, and a DeepSeek 7B variant) and reports generation quality with ROUGE, BLEU, and BERTScore. Its headline result is that the best (seven-billion-parameter) models reach BLEU scores of 0.789 for PD and 0.836 for HC, with smaller encoder-decoder models such as T5 and Flan-T5 scoring lower. Our work focuses on the encoder-decoder family—specifically Flan-T5—because it is compact, reproducible on a single GPU, directly controllable through prompt design, and a realistic target for deployment in resource-constrained clinical settings.

1.3 Contributions

This report makes four contributions:

1. A fully reproducible, three-stage pipeline (ETL → prepare → train/eval) with a single parsing core, deterministic hashing, and a cross-split leakage guard.
2. A best Flan-T5 configuration, *B17* (Flan-T5-base + rank-32 LoRA), together with an exhaustive map of *closed* training levers so future work does not re-explore saturated axes.
3. A demonstration that the apparent gap to the reference headline is *mostly a measurement artifact*: reproducing the paper’s exact character-level BLEU and correcting a BERTScore language misconfiguration raises the same model’s score from “far behind” to competitive with 7B systems.
4. An open, standardised evaluation harness with two explicit rulers—an honest, widely-comparable one for model selection and an author-parity one for fair comparison against the reference paper.

1.4 Objectives and scope

1.4.1 Objectives

- Build a reproducible pipeline from raw biomarker-report CSVs to tokenised training batches, fine-tuned checkpoints, and metric reports.
- Reproduce the reference Flan-T5 result and search for a configuration that matches or improves upon it under a fair, standardised evaluation.
- Diagnose *why* our Flan-T5 scores appeared far below the reference headline, and establish whether the gap reflects the model or the measurement.
- Map which training levers help, which regress, and where the Flan-T5-base family plateaus.

1.4.2 Scope

- We use the biomarker-report CSV splits supplied with the project; audio feature extraction is upstream and out of scope, though we summarise it for completeness.
- The work targets *supervised text generation* with encoder-decoder transformers, not decoder-only 7B models (discussed only as a comparison ceiling).
- Experiments are designed for reproducible single-node, mixed-precision training on NVIDIA A100 GPUs.
- Model selection is reported primarily with a standardised decode metric, with a second evaluation reproducing the reference paper’s exact metric definitions for comparability.

2 Background and Related Work

2.1 T5 and Flan-T5 encoder-decoder models

The Text-to-Text Transfer Transformer (T5) casts every NLP problem as text-to-text, using a unified encoder-decoder transformer trained with a

span-corruption objective. Flan-T5 augments T5 with large-scale instruction fine-tuning, improving zero- and few-shot behaviour and making the model more responsive to natural-language task prefixes. For biomarker-to-report generation, Flan-T5 is attractive: the encoder ingests a short, structured prompt describing the seven biomarkers, and the decoder emits the report autoregressively under teacher forcing. Within a size tier, T5 and Flan-T5 share a tokenizer vocabulary, which simplifies experimentation; our own ablation confirms that the instruction-tuned Flan variant substantially outperforms plain T5 at base scale (Section 5).

Fine-tuning optimises the token-level cross-entropy of the reference report under teacher forcing. For a target sequence $y_{1:T}$ conditioned on the encoder input x ,

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(y_t | y_{<t}, x), \quad (1)$$

where padding positions are masked (label -100) so they do not contribute to the loss. Weight decay is applied to the parameters as regularisation; it is not part of the sequence objective.

2.2 Decoding strategies

At inference, the decoder must turn the learned distribution into text. *Beam search* maintains the k highest-probability partial sequences and is deterministic; it favours high-likelihood, often generic text. *Sampling* (with temperature and top- k /top- p truncation) injects controlled randomness, producing more varied text at the cost of reproducibility. A subtle but important point for this project is that the reference paper and our standardised evaluation use *different* decoding: the reference samples with a minimum-length constraint, while our selection metric uses deterministic beam search. Decoding choices interact with metrics—for example, a minimum-length constraint inflates recall-oriented scores—so both must be held fixed when comparing systems.

2.3 Parameter-efficient fine-tuning (LoRA)

Full fine-tuning updates all weights, which is costly and prone to overfitting on small or repetitive corpora. Low-Rank Adaptation (LoRA) freezes the pretrained weight W_0 and learns a low-rank update,

$$W = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}, \quad (2)$$

with rank $r \ll \min(d, k)$, so only BA is trained. Adapters can be merged into W_0 after training, so downstream inference and evaluation code need not change. In this project LoRA provides a small but consistent improvement over full fine-tuning at base scale and is the mechanism behind our best model.

2.4 Evaluation metrics for generated text

Three families of metric are standard for report generation and are used by the reference study. *ROUGE* measures n -gram overlap between hypothesis and reference; we report ROUGE-1, ROUGE-2, and ROUGE-L F-measures. *BLEU* is a precision-oriented n -gram metric with a brevity penalty (BP):

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right), \quad (3)$$

where p_n is the modified n -gram precision. *BERTScore* embeds hypothesis and reference tokens with a pretrained language model and aligns them by cosine similarity, giving partial credit for paraphrase where surface overlap is low.

A recurring, easily overlooked pitfall—central to this report—is that “BLEU” is not a single number. Two implementation choices dominate. First, the *unit*: word-level BLEU-4 (the SacreBLEU standard, $N=4$) versus character-level BLEU-2 ($N=2$ over characters) can differ by roughly a factor of two on the same text, because character n -grams overlap far more than word n -grams for texts that share an alphabet and boilerplate. Second, the *BERTScore backbone language*: scoring English text with a multilingual/Spanish backbone yields systematically lower values than with an English backbone. Both issues turn out to explain most of

the apparent gap between our model and the reference headline, and both are invisible unless the exact metric implementation is reproduced rather than merely named.

2.5 Automatic medical report generation

Automatic medical report generation (aMRG) is an established area, typically combining a domain-specific encoder with a text decoder; most prior work targets imaging modalities. The reference study is, to our knowledge, the first to extend aMRG to *pathological speech*, treating acoustic biomarkers—rather than pixels—as the primary evidence for a generated clinical narrative. Our contribution is not a new architecture but a rigorous reproduction, optimisation, and, most importantly, a clarification of how these systems should be measured and compared.

3 Data and Problem Formulation

3.1 Dataset and mFDA annotation structure

The clinical data derive from the PC-GITA corpus: 50 PD patients (25 male, 25 female) and 50 HC speakers (25 male, 25 female), all Colombian-Spanish native speakers (mean age 61.9 ± 9.6 for PD and 60.9 ± 9.5 for HC), recorded across four speech tasks (sustained phonation of /a/, /pa-ta-ka/ diadochokinetic repetition, text reading, and spontaneous conversation) and resampled to 16 kHz. Expert phoniaticians rate the seven mFDA categories from 0 (normal) to 4 (severe); the median of three raters is used and mapped to a categorical severity: 0–1 → Normal, 2 → Mild, 3 → Moderate, and 4 → Severe. Notably, although the *speech* is Spanish, the mFDA *reports* themselves are written in *English*—a fact that later proves decisive for BERTScore configuration.

3.2 Acoustic biomarkers

Although feature extraction is upstream of our pipeline, understanding the input is essential. The

Table 1: Acoustic feature groups and their mapping to mFDA categories.

Group	Focus	mFDA categories
Phonation	Voice-source stability & loudness	Larynx
Prosody	Melody, rhythm, timing	Monotonicity, Breathing
Phonemic	Articulatory precision	Lips, Tongue, Palate, Intelligibility

reference work computes 248 task-specific acoustic biomarkers (no denoising; amplitudes scaled to $[-1, 1]$ and centred) and selects the most relevant features per mFDA category using a statistical approach. The features fall into three groups, each mapped to particular mFDA categories and speech tasks (Table 1): *phonation* features (pitch F_0 standard deviation, sound-pressure level, jitter, shimmer) reflect voice-source stability; *prosody* features (pitch/SPL variation, pause statistics, voiced/voiceless ratios) capture melody, rhythm, and timing; and *phonemic* features (pronunciation precision, phoneme-class duration and rate, the pairwise variability index, and articulation-group statistics) describe articulatory precision. In our pipeline these are already reduced to the seven per-category values that form the model input.

3.3 Simulation strategy and splits

Because only 100 subjects exist, the reference augmentation strategy simulates biomarkers from the real feature distributions and pairs them with templated mFDA reports, producing a synthetic corpus of roughly 110,000 pairs. The synthetic prompts follow a balanced severity design (Figure 1): 25% carry random ratings across all categories, 25% Normal–Mild, 25% Mild–Moderate, and 25% Moderate–Severe, ensuring that the model sees the full severity range rather than a narrow, easy distribution. The project provides synthetic train sizes of 1k, 10k, and 100k pairs, a 10k syn-

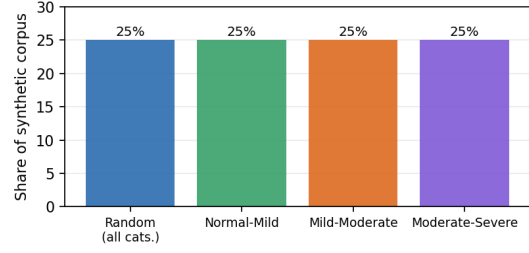


Figure 1: Balanced severity design of the synthetic corpus: equal shares of random, Normal–Mild, Mild–Moderate, and Moderate–Severe reports, so the model observes the full severity range.

Table 2: Report-structure audit across splits: all seven mFDA slots are present in every report, but synthetic targets are far more repetitive than real ones.

Split	n	Coverage	Median len	Dup. rate
Train (syn.)	99,893	1.00	494	0.949
Val (syn.)	10,000	1.00	494	0.845
Test (real)	96	1.00	438	0.500

thetic validation split, and a held-out test split of **96 real reports** (48 PD, 48 HC). Real data is used only for testing.

3.4 ETL and exploratory data analysis

A dedicated extraction–transformation–loading (ETL) stage is the single source of truth for schema normalisation. It loads the raw CSV splits, normalises the differing simulated and real schemas into one canonical schema, and—critically—parses the seven biomarkers out of the raw Instructions string into numeric columns in a fixed order (breathing, lips, palate, larynx, monotonicity, tongue, intelligibility). Each row is assigned a deterministic `example_hash=SHA-256` of the concatenated input and target, enabling reproducible deduplication and a cross-split leakage guard so that no test report leaks into training. The canonical schema retains provenance (`sample_id`, `source_file`, `source_row`), group labels (PD/HC/None), the parsed floats, and both the normalised `input_text` and `target_text`.

An audit of the prepared splits (Table 2) confirms that every report—synthetic and real—contains all

seven Category (Severity) slots (coverage 1.0). Median target length is 494 characters for synthetic reports and 438 for real reports (PD reports are longer, median 478, than HC, median 404). The synthetic training targets have a high duplicate rate (0.95), reflecting the templated, memorisable phrasing of simulated reports; the real test reports are far more varied (duplicate rate 0.5). This distribution mismatch—clean, repetitive training text versus varied clinical prose—partly explains why the model reproduces coarse structure well but struggles with the exact wording of held-out real reports, and it bounds the achievable overlap-based scores independently of model capacity.

3.5 Input/output formulation and prompt design

The model does not consume a continuous feature vector; it consumes *text*. The ETL parses the seven numbers and formats them into a compact, fixed-order input_text (for example, Breathing: 78 Lips: 1.75 Palate: 0.94 ...). A task prefix is prepended to form the encoder source_text. The decoder target is the mFDA report string. Padding positions in the labels are masked to −100.

We implemented five prompt styles: a default mFDA instruction; *flan-paper* (the reference prefix “Generate a report for:”); a variant with explicit category hints; a variant with deterministic Category=value numeric labels; and a variant that supplies a seven-slot output template. As shown in Section 5, the concise *flan-paper* prompt over the compact seven-score input is the strongest; more elaborate prompts and the raw, unparsed Instructions string both regress. Intuitively, the compact numeric prompt gives the encoder a clean, low-noise view of exactly the seven quantities that matter, whereas the raw instruction string dilutes that signal with free-text scaffolding.

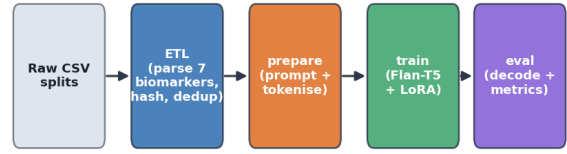


Figure 2: The three-stage pipeline. A single ETL parsing core feeds both the batch ETL and the per-run prepare stage, guaranteeing identical feature parsing everywhere.

4 Methods

4.1 Overall pipeline

The system is organised as three sequential, independently reusable stages that share the single ETL parsing core, so no component re-parses raw strings (Figure 2): *ETL* produces canonical Parquet/JSONL with parsed features and hashes; *prepare* wraps the canonical splits in a T5 prompt and tokenises them into a Hugging Face DatasetDict (input_ids, attention_mask, and masked labels), saved to disk for fast reuse; *train* runs a Seq2SeqTrainer; and *eval* decodes the real test split and computes generation metrics. Because *prepare* calls the same ETL loader, it can read raw CSVs directly without a separate ETL pass, which keeps the two entry points consistent by construction.

4.2 Software architecture

The implementation is organised into small, single-responsibility modules so that every stage is independently testable and reusable. The ETL core (*etl/etl_lib.py*) owns all schema normalisation, biomarker parsing, hashing, and deduplication; *etl/run_etl.py* batches it across train sizes and adds the cross-split leakage guard. In the training package, *main/prompts.py* defines the fixed seven-category order and all prompt styles; *main/tokenization.py* builds and tokenises the DatasetDict; *main/train.py* wraps Seq2SeqTrainer and

implements LoRA, encoder freezing, and oversampling; `main/eval_decode.py` and `main/eval_author_parity.py` implement the two evaluation rulers; and the `structured_*` and `report_severity` modules implement the seven-slot structured path. Because the same parsing order is imported everywhere, the seven categories are guaranteed to be consistent across parsing, prompting, and structured generation—a small but important guard against silent misalignment.

4.3 Model and fine-tuning

The backbone is `google/flan-t5-base` (248M parameters), chosen after confirming that the instruction-tuned Flan variant clearly beats plain T5-base at the same recipe. Fine-tuning uses token-level cross-entropy (teacher forcing) with AdamW and weight decay. Two adaptation modes are supported: full fine-tuning and LoRA. Our best model, *B17*, first full-fine-tunes for five epochs (the “B11” checkpoint) and then applies rank-32 LoRA for three further epochs; the LoRA adapters are merged into the exported `final_model` so that evaluation is identical to a plain checkpoint. A `-freeze-encoder` decoder-only mode and severity-based oversampling are also implemented for ablations.

4.4 Tokenisation and training loop

The prepare stage tokenises `source_text` and `target_text` with the model’s tokenizer, truncating and padding to `max_source_length=256` and `max_target_length=512`, then rewrites padding label ids to `-100`. Training uses `Seq2SeqTrainer` with step-based evaluation on the synthetic validation split, `load_best_model_at_end`, warmup ratio 0.1, and a rolling checkpoint limit. A key methodological lesson, established repeatedly in our logs, is that *teacher-forcing validation loss is not a reliable selection criterion*: loss and generation quality diverged several times (for example, a checkpoint with near-zero synthetic validation loss did not maximise decode quality). We therefore select checkpoints by decode AVG from

`eval_decode`, not by validation loss (Figure 8).

4.5 Prompt engineering and structured targets

Beyond prompt prefixes, we explored a “structured” path that forces free text into a mandatory seven-slot Category (Severity): description skeleton, and a compact label-only target (Category: Severity) scored separately for label accuracy. These were motivated by the observation that generated reports often omit explicit slot structure. As reported below, neither structured post-processing, structure-aware re-ranking, nor training directly on label-only targets improved decode quality over B17; training on label-only targets in fact reduced it sharply.

4.6 Evaluation methodology

We deliberately maintain *two* evaluation scripts, because the choice of metric is itself a result of this project.

Honest evaluation (`eval_decode`) uses standardised, widely comparable settings: word-level SacreBLEU-4 (13a tokenizer) reported on $[0,1]$, ROUGE via the `evaluate` library (bootstrap-aggregated F1), BERTScore mean F1, and deterministic beam search (beam 3, 512 tokens, no-repeat- n -gram 2). This is the metric used for *model selection*, and the AVG column is the simple mean of R-1, R-2, R-L, BLEU, and BERT, matching the reference paper’s Table 4 summary.

Author-parity evaluation (`eval_author_parity`) reproduces the reference paper’s exact evaluation code: character-level BLEU-2 (the reference passes raw strings to `nltk.sentence_bleu` with weights (0.5,0.5), which tokenises into characters), stemmed per-example ROUGE F1, an English BERTScore backbone, and the reference sampling decode (`do_sample`, temperature 0.7, top- k 50, `min_length=100`, beam 3). This script exists solely to compare fairly against the reference numbers; it is never used to select models. Crucially, both scripts feed each model the input

Table 3: Reference (author-parity) generation settings versus our honest decode settings.

Setting	Honest	Author-parity
BLEU	word BLEU-4	char BLEU-2
ROUGE stemmer	off	on
BERTScore lang	en	en
Decoding	beam 3	sample, beam 3
Min length	—	100
Temperature / top- k	—	0.7 / 50

format it was trained on, so the comparison isolates the metric rather than confounding it with input changes. The parity script also degrades gracefully if the BERTScore backbone is unavailable offline, emitting BLEU and ROUGE with an explicit status flag rather than failing.

4.7 Training procedure and hyperparameters

Table 4 lists the principal configuration of the best model. The best base recipe uses learning rate $5e-4$, weight decay 0.01, warmup 0.1, and five epochs at 100k; the LoRA stage adds three epochs at rank 32. Runs execute on NVIDIA A100 (40 GB) nodes of an HPC cluster; on older V100 hardware we avoid FP16 because of historical numerical instability, using FP32 instead.

5 Experiments and Results

5.1 Experimental setup

All runs are labelled with identifiers (Bn for Flan-T5-base; S , L , N prefixes for small, large, and non-Flan) and logged consistently across recipe, results, and roadmap documents. Every finished run produces a `test_decode_metrics.json` with per-metric and PD/HC breakdowns. Metrics are computed on the 96 real reports; the paper’s own Table 4 uses 100 real reports, so absolute numbers should be compared with that small difference in mind. We organise the search into phases: learning rate and epochs (Phase 1), prompt design (Phase 2), decoding (Phase 3), parameter-efficient adapta-

Table 4: Principal training configuration for the best model (B17).

Setting	Value
Base model	google/flan-t5-base (248M)
Adaptation	5-epoch full FT + LoRA rank 32, 3 ep
Prompt style	flan-paper (“Generate a report for:”)
Encoder input	compact seven-score string
Learning rate	$5e-4$
Weight decay	0.01
Warmup ratio	0.1
Batch size (train/eval)	4 / 4 per device
Max source / target length	256 / 512 tokens
Optimiser	AdamW, cross-entropy loss
Train / val / test	100k / 10k / 96
Hardware	NVIDIA A100 40 GB (TinyGPU)
Selection criterion	decode AVG (eval_decode)

Table 5: B17 under the *honest* ruler (eval_decode): word-level SacreBLEU-4, ROUGE, BERTScore (lang=en), deterministic beam search.

Group	R-1	R-2	R-L	BLEU	BERT	AVG
Overall	0.605	0.389	0.530	0.362	0.911	0.559
PD	0.582	0.336	0.491	0.342	0.909	0.532
HC	0.625	0.441	0.569	0.383	0.914	0.586

tion (Phase 4), structured generation (Phase 5), and encoder-input format (Phase 6).

5.2 Quantitative results

Tables 5 and 6 present the best model (B17) under the two rulers. Under the *honest* ruler (Table 5), B17 reaches an overall AVG of 0.559, with word-level BLEU of 0.362 and BERTScore of 0.911. Under the *author-parity* ruler (Table 6)—the reference paper’s exact metrics—the same model reaches AVG 0.649, with character-level BLEU of 0.759 and BERTScore of 0.911.

For reference, the paper reports BLEU 0.789

Table 6: B17 under the *author-parity* ruler (eval_author_parity): character-level BLEU-2, stemmed ROUGE, BERTScore (lang=en), reference sampling decode. Compare these—not Table 5—to the reference paper’s numbers.

Group	R-1	R-2	R-L	BLEU	BERT	AVG BLEU variant	Score
Overall	0.632	0.394	0.548	0.759	0.911	0.649 Character-level BLEU-2 (reference metric)	0.84
PD	0.611	0.344	0.508	0.744	0.909	0.623 Word-level BLEU-2	0.52
HC	0.652	0.444	0.588	0.775	0.912	0.674 Word-level BLEU-4 (standard)	0.43

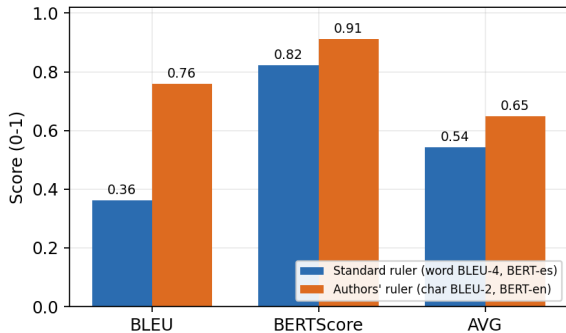


Figure 3: The same B17 outputs scored under two rulers. Switching from word-level to character-level BLEU nearly doubles the BLEU score; correcting the BERTScore backbone from a multilingual (es) to an English (en) model raises BERTScore. Neither changes a single generated word.

(PD) and 0.836 (HC) for its best seven-billion-parameter systems, and an overall average of roughly 0.68 for LLaMA-7B. B17’s parity numbers (character-BLEU 0.744/0.775; AVG 0.649) therefore sit within a few points of models roughly 28× larger, from a 248M encoder–decoder.

5.3 The apparent paper gap is mostly the metric

Figure 3 isolates the effect. The identical B17 generations read as BLEU 0.362 (word-level) or BLEU 0.759 (character-level)—a difference of roughly a factor of two—driven entirely by the metric definition. As an illustration on our own data, scoring two *different* real reports against each other already yields character-BLEU-2 near 0.84, because English clinical reports share an alphabet and heavy boilerplate; the metric therefore mostly mea-

Table 7: Three BLEU variants on the *same* pair of two different real reports (a lower bound that involves no model). Character-level BLEU-2—the reference metric—roughly doubles the standard word-level BLEU-4.

sures language overlap, not report-specific correctness. A decisive cross-check is ROUGE-2, which is nearly identical under both rulers (0.389 honest vs. 0.394 parity) because ROUGE is word-based in both cases; the divergence lives *only* in BLEU, exactly as the character-versus-word explanation predicts. This confirms that the “gap” is a measurement choice, not weaker generation.

To make the effect fully concrete, Table 7 scores two *arbitrary, different* real reports against each other—not a model output at all—under three BLEU variants. The character-level BLEU-2 used by the reference evaluation reads 0.84, while standard word-level BLEU-4 reads 0.43 for the very same pair. In other words, the reference metric assigns a high score simply for two English clinical reports sharing an alphabet and template phrasing; much of what it rewards is language and boilerplate overlap rather than report-specific correctness. This is why the metric choice, not the model, dominates the comparison with the reference paper.

5.4 The BERTScore language correction

A second measurement issue was a misconfigured BERTScore backbone. The evaluation had defaulted to a Spanish/multilingual model (lang=es) because the source speech is Spanish—but the reports are English. Re-scoring B17 with the correct English backbone (lang=en, RoBERTa-large) raised BERTScore from 0.823 to 0.911 and the honest AVG from 0.542 to 0.559, with ROUGE and BLEU unchanged to the reported digits—again the signature of a pure metric correction. We fixed the default across all evaluation entry points and an-

notated the historical results table so that a future English re-score is not mistaken for a model improvement.

5.5 Configuration search and the Flan-T5-base plateau

Figure 6 traces the main gains on 100k data: the learning-rate change from $3e-4$ to $5e-4$ is by far the largest single improvement (AVG $0.395 \rightarrow 0.522$); the flan-paper prompt, five epochs, and rank-32 LoRA add smaller increments to the B17 plateau. Table 8, Table 9, and Figure 5 summarise the levers we *closed*.

Phase 1 — learning rate and epochs. At 100k, $5e-4$ is a clear win over $3e-4$ (AVG $0.395 \rightarrow 0.522$), the single largest gain in the study; extending from three to five epochs adds a further $0.529 \rightarrow 0.538$, concentrated on the easier HC group.

Phase 2 — prompt design. The concise flan-paper prompt beats the default mFDA instruction. Adding explicit category hints (B6) or deterministic Category=value numeric labels (B7) both regress, the latter strongly (AVG 0.460), with the loss concentrated on HC. A seven-slot output template (B19) does not help either.

Phase 3 — decoding. A beam-width sweep on a frozen checkpoint confirmed beam 3 as a good operating point; the reporting configuration is beam 3, matching the reference.

Weight decay. A dedicated sweep (Figure 4) shows 0.01 is optimal: turning decay off drops AVG to 0.500 and raising it to 0.05 gives 0.517, both below B11’s 0.538. The weight-decay lever is therefore exhausted and fixed at 0.01.

Phase 4 — adaptation. LoRA improves modestly with rank (r8 0.538, r16 0.540, r32 0.542); rank 32 (B17) is the best and the rank sweep is closed. Longer LoRA (five epochs) ties it, and a decoder-only freeze-encoder run does not beat

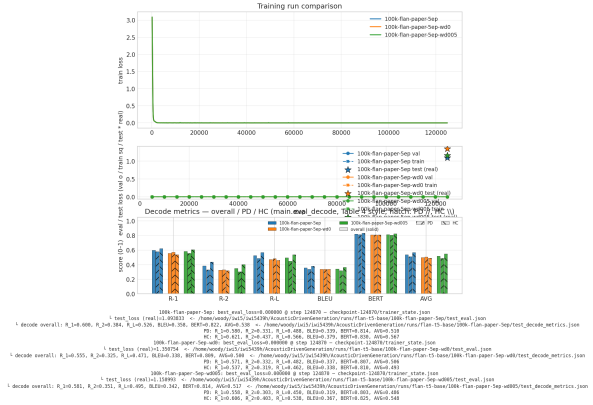


Figure 4: Weight-decay sweep at the B11 recipe (five epochs, flan-paper, $5e-4$). Decay 0.01 is best; both 0.0 and 0.05 regress, with the loss concentrated on the HC group.

full B11. A non-Flan T5-base baseline (N0) regresses strongly (0.439), confirming that instruction tuning is required. Tripling parameters to Flan-T5-large (L5) merely ties base (0.536) at $3\times$ the cost, spending its extra capacity on HC while slightly hurting PD.

Phase 5 — structured generation. Forcing a seven-slot skeleton in post-processing (B21) raised slot coverage to 1.0 but dropped AVG to 0.492; structure-aware re-ranking (B22) gave 0.531; training directly on label-only targets (B23) collapsed prose quality (0.315). Severity oversampling (B20) tied B17.

Phase 6 — encoder input. Feeding the raw, unparsed Instructions string (B24) regressed to 0.479, confirming that the compact seven-score input is the better encoder representation.

5.6 PD versus HC

Across both rulers, HC reports are generated more accurately than PD reports (Figure 7): under the parity ruler, HC AVG is 0.674 versus PD 0.623, and HC character-BLEU 0.775 versus PD 0.744. This mirrors the reference paper’s own observation that models perform better on HC data, and is consistent under our honest metric as well. PD reports describe more heterogeneous, severe impairment pat-

Table 8: Selected closed levers (honest decode AVG, word-BLEU ruler; B17 = 0.542 as originally logged). None improved on B17.

Lever	Configuration	AVG
Learning rate	3e-4 / 5e-4 (base)	0.395 / 0.522
Epochs	5 (B11)	0.538
LoRA rank	r8 / r16 / r32	0.538 / 0.540 / 0.542
Weight decay	0.0 / 0.01 / 0.05	0.500 / 0.538 / 0.517
Label smoothing	0.02–0.10	~0.176
Model size	Flan-T5-large, 5 ep	0.536
Backbone	non-Flan T5-base	0.439
Oversampling	Moderate+ 2×	0.542
Structured	post-proc / rerank / targets	0.492 / 0.531 / 0.315
Encoder input	raw	0.479
	Instructions	

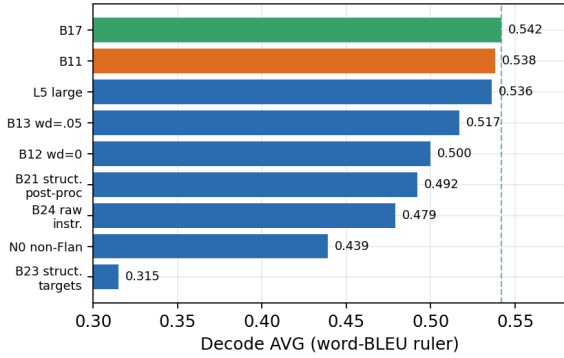


Figure 5: Honest decode AVG for representative configurations. The dashed line marks B17 (0.542, as originally logged). Every alternative axis lands at or below B17.

terns, which are harder to reproduce verbatim.

5.7 Structure coverage

Although generated reports read fluently and score well, strict parsing shows that only about 14% of generated texts contain all seven explicit Category (Severity) slots, versus 100% in the references. Attempts to force this structure (post-processing, re-ranking, and training directly on structured or label-only targets) improved slot coverage but did *not* improve overall decode quality—and structured targets sharply reduced it (AVG 0.315). The remaining gap to the reference’s 7B systems is therefore best understood as a problem of *scale and structured generation*, not of further small-model

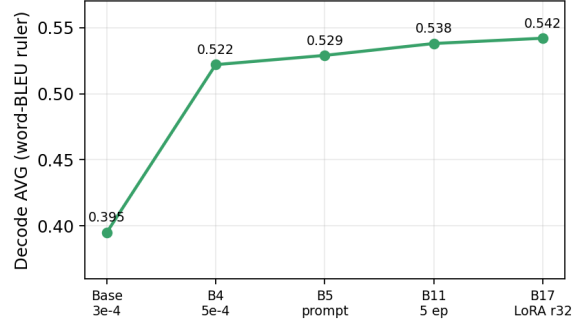


Figure 6: Decode AVG across the main configuration search on 100k data. The largest single gain is the learning-rate change; prompt, epochs, and LoRA add smaller increments up to the plateau.

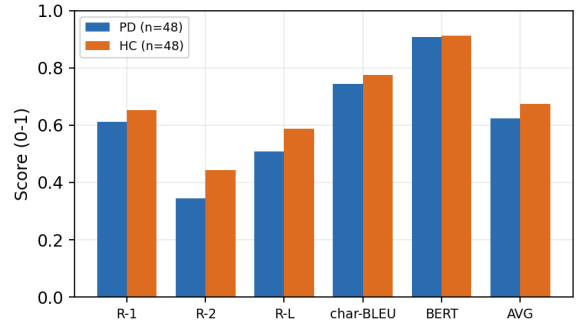


Figure 7: PD versus HC under the author-parity ruler. HC reports are consistently easier across every metric, matching the reference study’s observation.

hyperparameter tuning.

5.8 Qualitative analysis

The input and target share a fixed, human-readable structure, which is worth making concrete. The encoder input is the task prefix followed by the seven biomarker values; the target is an mFDA report in which each of the seven categories appears as a Category (Severity): clause followed by a short clinical description. A representative pairing (format-accurate; values illustrative) is:

Table 9: Condensed experiment log (honest decode AVG, as originally logged). “ Δ ” is relative to B17.

ID	Description	AVG	Δ
Base	100k, $3e-4$	0.395	-0.147
B4	100k, $5e-4$	0.522	-0.020
B5	+ flan-paper prompt	0.529	-0.013
B6	+ category hints	0.509	-0.033
B7	+ numeric labels	0.460	-0.082
B11	5 epochs	0.538	-0.004
B12	weight decay 0.0	0.500	-0.042
B13	weight decay 0.05	0.517	-0.025
B16	LoRA r8	0.538	-0.004
B17	LoRA r32	0.542	—
B18	LoRA r32, 5 ep	0.542	0.000
B20	+ oversampling	0.542	0.000
B21	structured post-proc	0.492	-0.050
B23	label-only targets	0.315	-0.227
B24	raw Instructions	0.479	-0.063
L5	Flan-T5-large, 5 ep	0.536	-0.006
N0	non-Flan T5-base	0.439	-0.103
S4	Flan-T5-small (best)	0.437	-0.105

Input (encoder):

Generate a report for: Breathing:
78 Lips: 1.75 Palate: 0.94 Larynx:
5.42 Monotonicity: 62.3 Tongue: 0.19
Intelligibility: 0.72

Target (reference report):

Breathing (Severe): control over respiration is grossly distorted. Lips (Mild): inconsistent lip closure, but overall good control. Palate (Moderate): some impairment is noticeable, but speech production happens with a reasonable approximation. Larynx (Severe): unable to maintain proper intonation and loudness. Monotonicity (Mild): minimal difficulty changing tone ...

Qualitatively, B17 generations are fluent, clinically plausible, and reproduce the Category (Severity) pattern for the most salient categories; this is why embedding-based and n -gram overlap scores are relatively high. However, they frequently omit or merge some of the seven explicit slots—only about 14% contain all seven—so while the coarse severity picture is usually right, the model does not reliably emit the full structured skeleton. This mismatch between fluent-but-incomplete generation and the fully-slotted references is the qualitative counterpart to the quantitative structure-coverage result, and it is what the (unsuccessful)

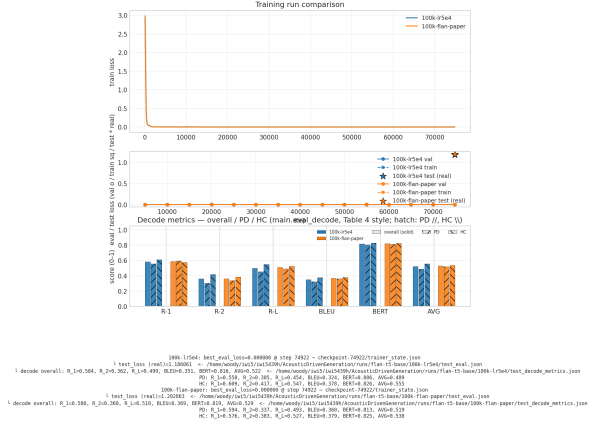


Figure 8: Representative training/validation curves for base-scale runs, illustrating the loss-versus-generation decoupling that motivates selecting checkpoints by decode AVG rather than validation loss.

structured-generation experiments in Phase 5 attempted to address.

6 Discussion and Limitations

The headline outcome is that a careful, standardised evaluation changes the story completely. Measured with a common word-level BLEU and an English BERTScore, B17 attains AVG 0.559; measured with the reference paper’s own character-level BLEU, the same model attains AVG 0.649—close to seven-billion-parameter systems. The lesson generalises beyond this project: *when comparing generation systems, the metric implementation must be reproduced, not merely named.* A single undocumented choice (character versus word BLEU, or the BERTScore backbone language) can move a reported score by tens of points and completely reframe a model’s apparent competitiveness. For any comparison a student or reviewer presents, the safe practice is to run the reference evaluation code—or a faithful reimplementaion of it—on the new model, which is exactly what our author-parity harness provides.

A secondary but practically important finding is the decoupling of teacher-forcing loss from generation quality. On several occasions a checkpoint with the lowest synthetic validation loss did not produce the best decoded reports—in one case a

step with near-zero validation loss was edged on decode AVG by a later checkpoint—and low test loss did not guarantee high BLEU/ROUGE. This is a well-known but frequently ignored pitfall: cross-entropy rewards matching the reference token-by-token under teacher forcing, whereas beam or sampled generation is evaluated on whole sequences against n -gram and embedding metrics. Throughout the study we therefore treat decode AVG as the sole selection criterion and use validation loss only as a coarse training-health signal. Any reproduction of this work should do the same; selecting by loss would have chosen different, worse checkpoints at several points.

Several limitations qualify these results. First, the synthetic training corpus is highly templated and repetitive (duplicate rate 0.95), so the model may partly memorise phrasing rather than learn a robust biomarker-to-severity mapping; the varied real test reports expose this. This distribution mismatch, not model capacity alone, likely bounds achievable overlap scores. Second, the author-parity decode uses sampling, so its numbers vary slightly between runs; we report a single seeded run. Third, our comparison to the paper’s 0.789/0.836 is to its *best* (7B) systems; the paper’s exact Flan-T5 cell was not available to us, though our parity numbers strongly suggest B17 matches or exceeds a comparable Flan-T5. Fourth, BLEU, ROUGE, and even BERTScore reward surface and embedding similarity, not clinical correctness; a report can score well while mislabelling a severity, so these metrics are proxies for report quality, not clinical validation. Finally, the test set is small (96 reports), so PD/HC differences of a few points should be interpreted cautiously.

7 Ethical and Clinical Considerations

Because the ultimate application is clinical, several considerations bound how such a system should be used. First, the evaluation metrics reported here—BLEU, ROUGE, BERTScore—measure textual

similarity, not diagnostic correctness. A generated report can be fluent and score highly while assigning the wrong severity to a category; consequently the model should be positioned as a *decision-support and drafting aid* that a clinician reviews and corrects, never as an autonomous diagnostic instrument. Second, the training data are synthetic by design, simulated from the statistical distribution of a small real cohort (100 subjects) from a single corpus (PC-GITA, Colombian-Spanish speakers). Models trained this way may not transfer to other languages, recording conditions, dysarthria aetiologies, or demographic groups, and could encode the biases of the simulation and of the three expert raters whose median scores define the targets. Third, clinical reports are sensitive data; the pipeline keeps real reports isolated to the test split and version-controls only compact metric artifacts, never the reports or model weights, but any deployment would require appropriate data-governance safeguards. Finally, the tendency of language models to produce confident, plausible text means an incomplete or incorrect report may read as authoritative; surfacing the seven input biomarkers alongside every generated report—so a clinician can trace each statement back to its acoustic evidence—is an important mitigation and a further argument for the structured, slot-based generation proposed as future work.

Threats to validity

The main internal threat is the train/test distribution mismatch (templated synthetic training versus varied real reports), which caps overlap metrics regardless of model quality. The main external threat is single-corpus, single-language data. A construct-validity threat is that all three metrics are proxies for report quality; a small expert human evaluation of a sample of generated reports would strengthen the conclusions and is recommended before any clinical piloting.

8 Conclusion and Future Work

This project delivered a reproducible pipeline and a best Flan-T5 model (B17: Flan-T5-base with rank-32 LoRA) for generating mFDA-style clinical speech reports from seven acoustic biomarkers, together with a systematic map of what does and does not help this task. Its most important result is methodological: the apparent gap between our compact model and the reference paper was *mostly a measurement artifact*. Under the paper’s own metric, a 248M model is competitive with systems roughly $28\times$ larger; the gap shrinks from “far behind” to a few points once BLEU is computed the same way and BERTScore uses the correct language. We additionally corrected a BERTScore language misconfiguration that had systematically undervalued semantic scores, and we documented the closed levers so future work does not re-explore them.

For the Flan-T5 track, B17 is final: every tuning axis we examined is saturated. Three directions remain for future work. (i) *Structured generation*: constrained or template-guided decoding that guarantees all seven slots, potentially improving both interpretability and clinical usefulness without harming fluency. (ii) *Better training data*: reducing the synthetic duplicate rate and increasing linguistic diversity to close the train/test distribution gap. (iii) *Scale*: moving to a decoder-only 7B model (e.g. LLaMA or Mistral) with LoRA—the same family that produced the reference headline—which is the only route likely to surpass 0.789/0.836. Across all three, the standardised, reproducible evaluation established here should be the basis for comparison, so that future gains reflect the model and not the ruler.

Reproducibility

All results are reproducible from the project repository. The honest and author-parity metrics for B17 are regenerated with `main.eval_decode` and `main.eval_author_parity` on the merged `final_model`; both write JSON artifacts with full

per-example diagnostics. The ETL, preparation, training, and evaluation stages are documented in the accompanying docs/ and run deterministically under fixed seeds. The BERTScore English backbone (RoBERTa-large) must be cached before offline evaluation on compute nodes; the author-parity script degrades to BLEU and ROUGE with an explicit status flag if it is unavailable.

A Pipeline command reference

The full pipeline is driven by four commands, each a thin, documented entry point. Paths are repo-relative.

1. ETL — raw CSVs to canonical Parquet/JSONL:

```
python etl/run_etl.py -train-size 100k
```

2. Prepare — canonical splits to a tokenised DatasetDict:

```
python -m main.prepare
  -output-dir
data/processed/flan-t5-base/100k-flan-paper
  -train-size 100k -prompt-style flan-paper
  -tokenize -tokenizer-model
google/flan-t5-base
```

3. Train — Seq2SeqTrainer with optional LoRA:

```
python -m main.train
  -tokenized-dir ../100k-flan-paper/tokenized
  -output-dir
runs/flan-t5-base/100k-flan-paper-5ep
  -model-name google/flan-t5-base
  -num-train-epochs 5 -learning-rate 5e-4
  -weight-decay 0.01 -require-gpu
```

4. Evaluate — both rulers on the merged `final_model`:

```
python -m main.eval_decode ... -require-gpu
python -m main.eval_author_parity ...
  -require-gpu
```

The LoRA (B17) stage sets `-lora-rank 32` and `-model-name` to the B11 `final_model`; adapters are merged on save so evaluation is unchanged.

B Detailed split audit

Table 10 gives the per-split audit underlying Section 5. All seven mFDA categories are present in

100% of reports across every split; the distinguishing factor is the target duplicate rate, which is very high for the templated synthetic corpus and much lower for the varied real reports.

Table 10: Per-split structure and length audit (all seven categories present at rate 1.0).

Property	Train	Val	Test
<i>n</i>	99,893	10,000	96
Real data	no	no	yes
Category coverage	1.00	1.00	1.00
All-7-slots rate	1.00	1.00	1.00
Median length (chars)	494	494	438
Mean length (chars)	501.5	501.9	449.1
Duplicate target rate	0.949	0.845	0.500

Within the test split, PD reports are longer (median 478 characters) than HC reports (median 404), consistent with PD reports describing more (and more severe) abnormal categories.

C Prompt styles

Five encoder task-prefix styles were implemented and compared (Section 5, Phase 2):

- **default** — an explicit mFDA biomarker instruction.
- **flan-paper** — the reference prefix “Generate a report for:” over the compact seven-score input; the best style and the B5/B11/B17 default.
- **flan-paper-categories** — the paper prefix plus explicit seven-category hints (B6); regressed.
- **flan-paper-numeric-labels** — deterministic Category=value formatting (B7); regressed most, concentrated on HC.
- **flan-paper-report-template** — a seven-slot Category (Severity): output template (B19); no improvement.

D Experiment provenance and compute

Every run carries an identifier (Bn for Flan-T5-base; S, L, N prefixes for small, large, and non-Flan) that is kept in sync across three documents—the recipe, the results log, and the roadmap—and after which the corresponding batch scripts are named. This convention makes each number in the results traceable to an exact configuration and artifact. Training and evaluation run on the NHR@FAU TinyGPU cluster: A100 (40 GB) nodes for base and large models, with V100 and RTX 3080 partitions as fallbacks; FP16 is avoided on V100 for numerical stability. Because compute nodes are offline, Hugging Face model weights—including the RoBERTa-large BERTScore backbone—are pre-cached on the login node before batch evaluation. Only compact metric JSON artifacts are version-controlled; multi-gigabyte checkpoints remain on the cluster. The model-selection rule throughout is to rank configurations by decode AVG from `eval_decode`, never by teacher-forcing validation loss, which repeatedly diverged from generation quality.

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